

Tools and APIs

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of tools and api contracts.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Conversation State and Memory. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 21: Tools and APIs

A tool call is a structured request to an external capability with returned data.



CREATOR MISSION

Match tool schemas to requests and handle one failure.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Prompt injection boundaries. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week’s end goal

By the end, I can explain tools and apis using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use tools and apis to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Tools expose structured operations with input and output contracts. Generating a tool call is a proposal; validation and authorisation decide whether it executes.

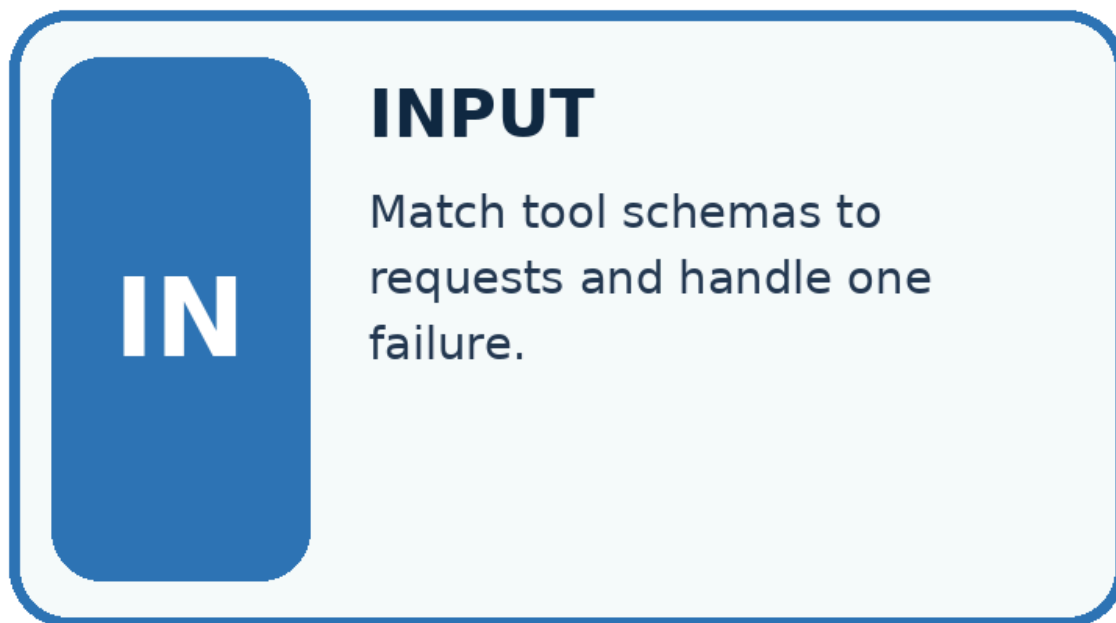
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

A tool call is a structured request to an external capability with returned data.



OUT

EVIDENCE

Match tool schemas to requests and handle one failure.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

For `lookup_room` accept exactly `room_id` as a nonempty string; return found plus capacity or a not-found result. It is read-only. Reject unknown arguments and do not replace missing values with invented facts.

Tools and API contracts — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Proposed call `lookup_room` has `room_id=B` and extra argument `book=true`.

Check or calculation: For `lookup_room` accept exactly `room_id` as a nonempty string; return found plus capacity or a not-found result. It is read-only. Reject unknown arguments and do not replace missing values with invented facts.

Conclusion and limit: Reject the extra field under the closed input contract. The lookup tool provides no booking authority even if the requested room exists.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Add `lend: true` to the same call.

Prediction: Reject the call before invoking the tool.

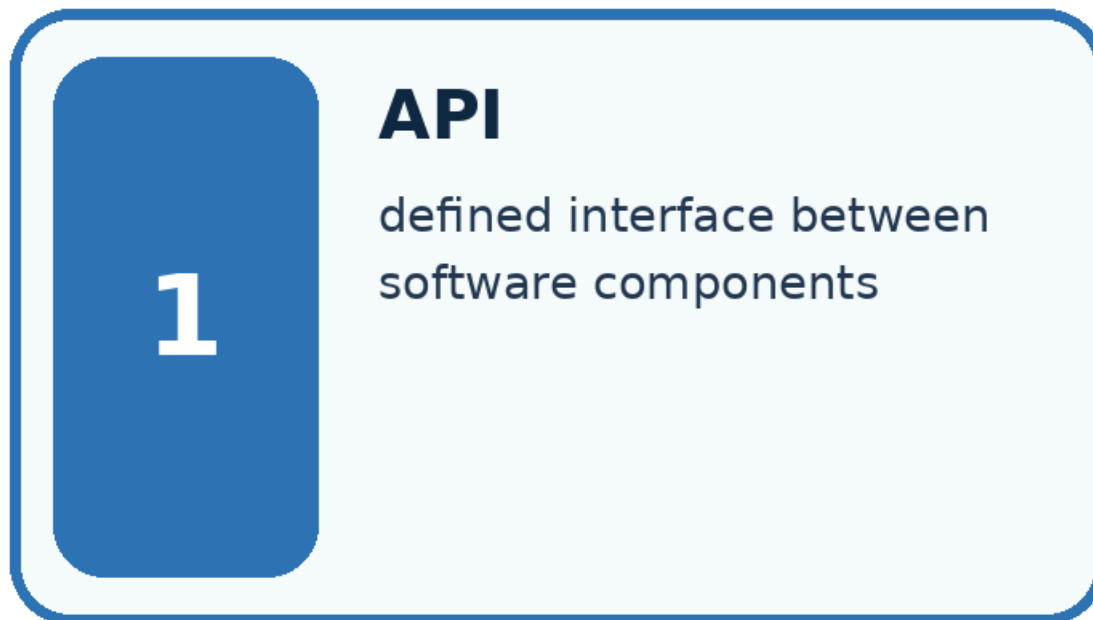
Reason: An extra operation is outside the exact allowed contract, even when `book_id` is valid.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

TOOL CALL

structured request to an external capability

3

ARGUMENT

named input sent to a tool

4

ERROR RESPONSE

structured explanation of failure

Word · My own example · Not this / boundary

Word: API

My own example: An agreed interface through which software requests data or actions from another system.

Not this / boundary: _____

Word: tool call

My own example: A structured request to a tool with specified arguments.

Not this / boundary: _____

Word: argument

My own example: A value supplied to a function or tool to tell it what to operate on.

Not this / boundary: _____

Word: error response

My own example: A message or structured result indicating that a request could not be completed as expected.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: API contract

Meaning: Declared input, output and failure rules for an interface

Word: Not found

Meaning: An explicit absence result rather than a fabricated value

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 21 • WORKED EVIDENCE

- 1 Tool schema requires operation and two numbers.
- 2 Request sends operation=divide, a=18, b=24.
- 3 Tool returns result=0.75; product formats 75%.
- 4 Timeout or invalid input must trigger retry, repair, or safe explanation—not invented data.

1. Tool schema requires operation and two numbers.
2. Request sends operation=divide, a=18, b=24.
3. Tool returns result=0.75; product formats 75%.
4. Timeout or invalid input must trigger retry, repair, or safe explanation—not invented data.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Proposed call `lookup_room` has `room_id=B` and extra argument `book=true`.

Reject the extra field under the closed input contract. The lookup tool provides no booking authority even if the requested room exists.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Tools and APIs

Check a library lookup call before it reaches a tool.

The facts we will use

`lookup_book` accepts exactly one field: `book_id`, a string. Registry IDs are C7 and M2. The proposed call is `{book_id: C7}`.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Tools and APIs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check shape

Exactly book_id is supplied, and C7 is text.



2. Check membership

C7 occurs in the authorised registry.



3. Read only

Run the lookup. This tool contract contains no lending or deletion operation.

CHANGE ONE THING

Add lend: true to the same call.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Check shape

What happens and why: Exactly book_id is supplied, and C7 is text.

Step: 2. Check membership

What happens and why: C7 occurs in the authorised registry.

Step: 3. Read only

What happens and why: Run the lookup. This tool contract contains no lending or deletion operation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A valid tool call can still return unavailable or outdated records. Validate the response and access rights separately.

Read the labelled picture: Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Add lend: true to the same call. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A model can use any tool just by describing the desired action. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project,

select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A successful tool result supports only its declared data and operation; it does not authorise subsequent side effects.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Tools and API contracts”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Tools and API contracts. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does a valid C7 ID permit the extra lending action? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Match requests to schemas, validate arguments, simulate successful and failed calls, and design a maximum retry rule.

Match requests to schemas, validate arguments, simulate successful and failed calls, and design a maximum retry rule.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Allow only retrieval and calculator tools with validated arguments and visible results.

THIS WEEK’S CHECKPOINT Allow only retrieval and calculator tools with validated arguments and visible results.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

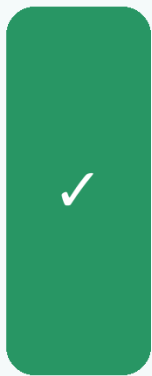
For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Match tool schemas to requests and handle one failure.

CHALLENGE Match tool schemas to requests and handle one failure.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

The question

Choose and verify tools.

Worked explanation

Calculator returns 108; search retrieves the definition. Verify inputs, outputs and supporting passage before answering.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Choose a tool for each subtask

Evidence or stage · Value or result

Evidence or stage: 12 times 9

Value or result: Calculator → 108

Evidence or stage: Habitat definition

Value or result: Search approved class pack

Evidence or stage: Before answering

Value or result: Verify input, result and support

An assistant has calculator and source-search tools. Request: 12 times 9 and habitat definition from a class pack.

Choose and verify tools.

Calculator returns 108; search retrieves the definition. Verify inputs, outputs and supporting passage before answering.

14. Try a new case independently

Independent case: Apply a new case independently

Tool accepts room_id string. Calls are {room_id:7}, {room_id:K} and {room_id:K, delete:true}. Validate; K is absent from the registry.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A tool can look up book availability and also place an order. The user asks only whether a book is available. Which action is authorized?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Allow only retrieval and calculator tools with validated arguments and visible results.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A tool can calculate or search supplied notes. The request is 'What is 17×6 , and where do roots take up water?' Assign each subtask and describe output checks.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Check a library lookup call before it reaches a tool.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of tools and api contracts

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of tools and api contracts

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check a library lookup call before it reaches a tool.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Run the lookup. This tool contract contains no lending or deletion operation.

1. Check shape
2. Read only
3. Check membership

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Read only Run the lookup. This tool contract contains no lending or deletion operation. Read the picture in order. Run the lookup. This tool contract contains no lending or deletion operation. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Add lend: true to the same call. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Reject the call before invoking the tool.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Reject the call before invoking the tool. An extra operation is outside the exact allowed contract, even when book_id is valid.

Repair a missing building block

Review Conversation State and Memory (ai-3/20). Goal: Explain the mechanism and terminology of conversation state and retention. Short practice: What is selected_book after the correction and after exit? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the mechanism and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Source Quality (ai-3/15). Goal: Explain the mechanism and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.